

A Ternary $(4k + 1(2))/3$ Collatz-Type Map with Two Attracting Cycles: Theoretical Analysis and Exhaustive Computational Verification up to 10^{11}

Farhad Banazadeh
fn.bana@hotmail.com
ORCID: 0009-0004-7023-0298

18 September 2026

Abstract

This paper studies the ternary $(4k + 1(2))/3$ Collatz-type map, based on division by 3 and a complete 3-adic compression step. The starting values are positive integers not divisible by 3. For $n = 3p + 1$ the pre-reduced value is $4p + 2$, while for $n = 3p + 2$ it is $4p + 3$. After this step, all factors of 3 are removed. In the compressed convention used in the computation, the state 3 is represented by 1. Two cycles are observed: $C_1 = (1, 2)$ and $C_7 = (7, 10, 14, 19, 26, 35, 47)$. A 3-adic heuristic based on Haar measure gives the distribution $\Pr(v_3 = k) = 2/3^{k+1}$ and $\mathbb{E}[v_3] = 1/2$. This gives the average logarithmic drift $\delta = \ln(4/3) - (1/2) \ln 3 \approx -0.2616$, corresponding to an average multiplicative factor of about 0.7698, or about a 23.02% decrease per step on the logarithmic model. This is a heuristic, not a proof of global convergence.

An exhaustive computation of every admissible starting value $1 \leq n \leq 10^{11}$ was completed by combining the previously audited range $1 \leq n \leq 10^9$ with a new exhaustive extension $10^9 < n \leq 10^{11}$. There are exactly 66,666,666,667 admissible starts. Every one was classified into one of the two observed cycles, with zero unknown cases, zero arithmetic overflows, and zero step-limit cases. The C_1 basin contains 2,179,160,439 starts (3.2687406585%), and the C_7 basin contains 64,487,506,228 starts (96.7312593415%). The combined computation used 5,449,566,633,079 steps to known attractors, with mean 81.7434994958 steps per admissible start. The maximum number of steps to a known attractor was 487, attained by $n = 81,078,368,921$. The largest value reached by any tested trajectory was 1,261,811,304,250,099,959,379, obtained from $n = 96,110,420,480$ at step 165. A maximum of 23 extra factors of 3 was removed in a single compressed step, and the combined total number of extra removed factors was 2,755,160,788,171. These results give strong finite computational evidence for the conjecture that every positive integer eventually enters one of the two cycles; they do not constitute a proof of global convergence.

1 Introduction

The classical Collatz problem asks about the long-term behavior of a very simple integer map. Many related maps replace the division by 2 with other divisors, use several residue classes, or compress an orbit by removing selected prime factors. The present work studies one such ternary system, built around division by 3 and complete removal of factors of 3 after each pre-reduced step.

The main difference from the earlier map studied by the author is important. The earlier map used the two branches $(4k - 1)/3$ and $(4k + 1)/3$ together with the direct $k/3$ branch. The present map instead uses $4k + 2$ for the residue class $1 \pmod{3}$ and $4k + 1$ for the residue class $2 \pmod{3}$, followed by complete 3-adic compression. The present system therefore has a different orbit structure and two observed terminal cycles.

The paper has three goals. First, it gives a precise definition of the map and explains why every step is integral. Second, it develops the 3-adic and average-logarithmic analysis supplied in the theoretical note. Third, it records the large computational experiment through 10^{11} , including basin sizes, step counts, peak values, factor-of-3 statistics, and record cases.

The computational evidence is deliberately separated from the conjecture. A finite exhaustive computation can establish the behavior of every tested starting value, but it cannot by itself establish the same statement for all positive integers.

2 Definition of the map

Let

$$D = \{n \in \mathbb{N} : 3 \nmid n\}.$$

Every $n \in D$ belongs to exactly one of the two residue classes

$$n = 3p + 1 \quad \text{or} \quad n = 3p + 2, \quad p \geq 0.$$

Define the pre-reduction map $G : D \rightarrow \mathbb{N}$ by

$$G(n) = \left\lfloor \frac{4n + 2}{3} \right\rfloor.$$

Equivalently,

$$G(3p + 1) = 4p + 2, \quad G(3p + 2) = 4p + 3.$$

The reason for the floor is only to write the two residue cases in one expression; the two displayed formulas show that $G(n)$ is always an integer.

For a positive integer m , let

$$v_3(m) = \max\{r \geq 0 : 3^r \mid m\}$$

be the exponent of 3 in m . The compressed map is

$$T(n) = \frac{G(n)}{3^{v_3(G(n))}}. \tag{1}$$

Thus $T(n)$ is not divisible by 3 after the compression.

The computational convention used in the exhaustive program identifies the eliminated state 3 with the state 1. In particular,

$$1 \longrightarrow 2 \longrightarrow 1.$$

This convention is part of the definition of the compressed state space used for the reported computation.

The sequence generated from an initial value $x_0 \in D$ is

$$x_{j+1} = T(x_j), \quad j \geq 0.$$

3 How the rule works: numerical examples

The map is easiest to understand through direct examples.

For $n = 1 = 3(0) + 1$,

$$G(1) = 4(0) + 2 = 2, \quad T(1) = 2.$$

For $n = 2 = 3(0) + 2$,

$$G(2) = 4(0) + 3 = 3 = 3^1, \quad T(2) = 1.$$

Hence $1 \rightarrow 2 \rightarrow 1$.

For $n = 4 = 3(1) + 1$,

$$G(4) = 6, \quad T(4) = 2,$$

so $4 \rightarrow 2 \rightarrow 1 \rightarrow 2 \rightarrow 1 \cdots$.

For $n = 5 = 3(1) + 2$,

$$G(5) = 7, \quad T(5) = 7.$$

Therefore 5 enters the seven-cycle immediately.

A central example is

$$7 \rightarrow 10 \rightarrow 14 \rightarrow 19 \rightarrow 26 \rightarrow 35 \rightarrow 47 \rightarrow 7. \quad (2)$$

Every step from 7 through 35 has no factor of 3 to remove. At 47,

$$G(47) = 4(15) + 3 = 63 = 3^2 \cdot 7,$$

and therefore

$$T(47) = 63/9 = 7.$$

This closes the cycle.

Another example showing compression is

$$2 \rightarrow 3 \rightarrow 1 \rightarrow 2 \rightarrow 1 \cdots,$$

where the state 3 is normalized to 1. The compression is important: it makes the dynamics live on the nonmultiples of 3 and records the complete 3-adic removal in a single step.

4 The two observed cycles

The first observed cycle is

$$C_1 = (1, 2),$$

with length 2. Its exact transitions are

$$1 \rightarrow 2, \quad 2 \rightarrow 1.$$

The second observed cycle is

$$C_7 = (7, 10, 14, 19, 26, 35, 47),$$

with length 7.

For the seven-cycle, the first six transitions are

$$7 \rightarrow 10 \rightarrow 14 \rightarrow 19 \rightarrow 26 \rightarrow 35 \rightarrow 47,$$

and the final transition is

$$47 \rightarrow 63 \rightarrow 7$$

in the uncompressed description, or simply $47 \rightarrow 7$ in the compressed map. Thus the final step removes exactly two factors of 3.

The theoretical note gives multiplicative cycle-scale estimates

$$\lambda_1 = \frac{(4/3)^2}{3} = \frac{16}{27} \approx 0.5926 < 1,$$

and

$$\lambda_2 = \frac{(4/3)^7}{3^2} = \frac{16384}{19683} \approx 0.8324 < 1.$$

These values are useful heuristic stability indicators for the asymptotic multiplicative model. They should not be confused with ordinary derivatives of the integer map, because the actual map contains the discrete valuation operation.

For comparison, the exact product of the actual ratios around C_1 is

$$\frac{2}{1} \cdot \frac{1}{3} = \frac{2}{3} < 1,$$

while the exact product around C_7 telescopes to 1 because the cycle returns to its starting value. The values $16/27$ and $16384/19683$ are therefore best understood as the theoretical-note's asymptotic scale estimates rather than exact cycle multipliers.

5 Stopping time and peak

For an initial value $x_0 = n$, write its orbit as

$$x_0, x_1, x_2, \dots, \quad x_{j+1} = T(x_j).$$

For the two-cycle system, a natural stopping time is the first index at which the orbit becomes smaller than its initial value:

$$\sigma(n) = \min\{j \geq 1 : x_j < n\}, \tag{3}$$

when such an index exists. This is the stopping-time definition used in the theoretical analysis.

For an orbit segment up to a stopping time, define the peak by

$$P_\sigma(n) = \max_{0 \leq j \leq \sigma(n)} x_j. \tag{4}$$

For the large 10^{11} computation, an additional global trajectory peak was tracked independently of this stopping-time definition. It is the maximum value reached before the trajectory was classified into one of the observed cycles.

The distinction matters. An orbit may have a relatively short stopping time but still make a very large temporary excursion. This is exactly what happens in the largest-peak record reported below.

6 Methodology

Two complementary approaches were used.

6.1 Theoretical 3-adic analysis

The first approach studies the distribution of the number of factors of 3 removed after the pre-reduction step. The analysis uses the Haar-measure heuristic on the ring of 3-adic integers $\mathbb{Z}_3$ and assumes that large orbit values sample residue classes in an approximately uniform way.

6.2 Exhaustive computation

The finite verification covers every integer

$$1 \leq n \leq 10^{11}, \quad 3 \nmid n.$$

The total number of admissible starts is

$$10^{11} - \left\lfloor \frac{10^{11}}{3} \right\rfloor = 66,666,666,667.$$

The original audit covered $1 \leq n \leq 10^9$. To preserve that completed computation, the extension scan covered only

$$1,000,000,001 \leq n \leq 100,000,000,000, \quad 3 \nmid n,$$

which contains exactly 66,000,000,000 admissible starts. The two aggregate data sets were then combined arithmetically.

The extension program uses a fast 64-bit hot path and automatically promotes a trajectory to `unsigned __int128` when the next arithmetic operation cannot safely be represented in 64 bits. This safeguard is essential: the final global peak is larger than the maximum unsigned 64-bit integer. The scan used four POSIX threads, dynamic chunks of 30,000,000 raw integers, a checkpoint CSV, an explicit unknown counter, overflow detection, and a safety limit of 1,000,000 steps. The extension completed with zero unknown cases, zero overflows, and zero step-limit cases.

The extension run took 30,700.091 seconds (about 8.53 hours) on the author's MacBook Pro. Its complete chunk log, summary, source code, and the earlier 10^9 audit materials are included in the computational package.

7 Theoretical analysis: average logarithmic contraction

For large n , the pre-reduced value has the approximate scale

$$G(n) \approx \frac{4}{3}n.$$

If

$$v = v_3(G(n)),$$

then after complete removal of factors of 3,

$$T(n) = \frac{G(n)}{3^v} \approx n \frac{4}{3^{v+1}}. \quad (5)$$

Therefore the logarithmic change is approximately

$$\ln \frac{T(n)}{n} \approx \ln \frac{4}{3} - v \ln 3. \quad (6)$$

7.1 Ergodic and Haar-measure viewpoint

The theoretical note treats the large-scale orbit as sampling residue classes in an approximately uniform way. In the 3-adic setting, the natural reference measure is Haar measure on $\mathbb{Z}_3$. This does not mean that a deterministic orbit is literally random; rather, it gives a useful ergodic model for the frequency of residue classes and therefore for the frequency of powers of 3 removed by the compression.

The key quantity is the 3-adic valuation of the pre-reduced value after the compulsory division by 3. Under the uniform-residue or Haar-measure heuristic in $\mathbb{Z}_3$,

$$\Pr(v = 0) = 1 - \frac{1}{3} = \frac{2}{3},$$

$$\Pr(v = 1) = \frac{1}{3} - \frac{1}{9} = \frac{2}{9},$$

and, in general,

$$\Pr(v = k) = \frac{2}{3^{k+1}}, \quad k \geq 0. \quad (7)$$

Indeed, $v = k$ means divisibility by 3^k but not by 3^{k+1} . The probability of divisibility by 3^k is 3^{-k} , and the difference between successive probabilities gives (7).

7.2 Expected number of removed factors

Using (7),

$$\mathbb{E}[v] = \sum_{k=0}^{\infty} k \frac{2}{3^{k+1}} = \frac{2}{9} \sum_{k=1}^{\infty} k \left(\frac{1}{3}\right)^{k-1}.$$

Since

$$\sum_{k=1}^{\infty} k r^{k-1} = \frac{1}{(1-r)^2}, \quad |r| < 1,$$

we obtain

$$\mathbb{E}[v] = \frac{2}{9} \frac{1}{(1-1/3)^2} = \frac{1}{2}. \quad (8)$$

Thus the heuristic predicts one half of a factor-of-3 exponent on average per step.

7.3 Logarithmic drift

Taking the expectation in (6),

$$\delta = \mathbb{E} \left[\ln \frac{T(n)}{n} \right] \approx \ln \frac{4}{3} - \mathbb{E}[v] \ln 3.$$

Using (8),

$$\boxed{\delta = \ln \frac{4}{3} - \frac{1}{2} \ln 3 = \ln \left(\frac{4}{3\sqrt{3}} \right)}. \quad (9)$$

Numerically,

$$\delta \approx -0.2616 < 0.$$

Therefore the corresponding geometric mean factor is

$$e^\delta = \frac{4}{3\sqrt{3}} \approx 0.7698. \quad (10)$$

In this model, the orbit size decreases by about

$$1 - 0.7698 \approx 0.2302,$$

or 23.02%, per step on the logarithmic average scale.

This is the central theoretical reason for expecting contraction. The pre-reduction step multiplies the size by roughly $4/3$, but the average removal of 3-adic factors contributes an additional factor of roughly $3^{-1/2}$. The combined average factor is below 1.

7.4 Average multiplicative factor

Equation (10) gives more than a negative logarithmic drift. It gives the geometric mean scale factor

$$\rho = \exp(\delta) = \frac{4}{3\sqrt{3}} \approx 0.7698. \quad (14)$$

Thus the theoretical model predicts

$$\frac{T(n)}{n} \quad \text{to have geometric mean approximately } 0.7698$$

for large states. Equivalently, the logarithmic size is expected to decrease by about 0.2616 per step, while the corresponding geometric size decreases by about 23.02% per step. This is the main contraction mechanism proposed in the theoretical analysis.

7.5 What the negative drift does and does not prove

The negative value in (9) is a strong heuristic indication of downward behavior, but it is not a proof of convergence of every orbit. The valuation distribution was obtained from an asymptotic uniformity assumption, while an actual orbit is deterministic and its successive residue classes are not independent random samples.

In particular, the statement that divergence has probability zero belongs to the probabilistic heuristic. It should not be stated as a theorem about every integer orbit. A rigorous proof would require a deterministic argument controlling all possible long-term residue patterns, or a different global argument.

8 Algebraic view of the seven-cycle

The seven-cycle is especially simple because the first six compressed steps have no factor of 3 removed. Its values satisfy

$$7 \mapsto 10 \mapsto 14 \mapsto 19 \mapsto 26 \mapsto 35 \mapsto 47.$$

The final pre-reduced value is

$$G(47) = 63 = 3^2 \cdot 7.$$

Therefore

$$T(47) = \frac{63}{3^2} = 7.$$

The cycle closes only because the final step removes two powers of 3 at once.

This gives a useful picture of the dynamics. The increasing branches can produce a long upward run, but a sufficiently divisible pre-reduced value can produce a strong downward jump. The seven-cycle is a compact example of this balance: six consecutive increases are followed by one compressed return.

9 Computational verification up to 10^{11}

The exhaustive verification classifies every admissible starting value through 10^{11} . Table 1 gives the combined result from the original 10^9 audit and the new extension scan.

Table 1: Complete computational result for $1 \leq n \leq 10^{11}$, $3 \nmid n$.	
Quantity	Value
Upper bound N	100,000,000,000
Admissible starts	66,666,666,667
Cycle $C_1 = (1, 2)$ count	2,179,160,439
Cycle C_1 percentage	3.2687406585%
Cycle C_7 count	64,487,506,228
Cycle C_7 percentage	96.7312593415%
Unknown	0
Total steps to known attractor	5,449,566,633,079
Mean steps to known attractor	81.7434994958
Maximum steps to known attractor	487
Start attaining maximum steps	81,078,368,921
Global peak	1,261,811,304,250,099,959,379
Start attaining global peak	96,110,420,480
Step of global peak	165
Total extra factors of 3 removed	2,755,160,788,171
Maximum extra factors in one step	23

As a consistency check,

$$2,179,160,439 + 64,487,506,228 = 66,666,666,667,$$

so the two cycle counts account for every admissible starting value. No tested start remained unresolved.

The new extension interval alone contained 66,000,000,000 admissible starts: 2,157,375,328 entered C_1 and 63,842,624,672 entered C_7 . Its total step count was 5,406,806,745,632 and its total count of extra removed factors of 3 was 2,733,476,950,658.

The basin proportions remain strongly unequal in this finite domain: about 3.269% of admissible starts enter C_1 and about 96.731% enter C_7 . This is a statement about the tested finite range, not a proof of an asymptotic basin density.

10 A one-million range stopping-time study

The theoretical note also contains a detailed stopping-time study over the first million integers not divisible by 3. There are 666,667 admissible starts in that range. The reported maximum stopping time is

$$143$$

at

$$n = 497,116,$$

with a peak of

$$410,683,162$$

within the stopping-time segment.

The ten leading stopping-time records reported in that study are shown below.

Table 2: Top stopping-time records in the 10^6 study.

Rank	Start n	$\sigma(n)$	Peak
1	497,116	143	410,683,162
2	662,822	142	410,683,162
3	883,763	141	410,683,162
4	749,897	129	2,687,932,606
5	629,449	127	263,207,404,847
6	839,266	126	263,207,404,847
7	832,280	121	2,687,932,606
8	981,244	120	40,062,776,062
9	360,637	119	517,652,138
10	999,863	118	2,687,932,606

The value 497,116 is the stopping-time champion in the one-million study. It survives for 143 steps before first becoming smaller than its starting value, while its orbit reaches about 410 million in the measured segment. This is a useful example of why stopping time and peak size are different quantities.

The one-million experiment reported approximately 3.31% of starts entering C_1 and 96.69% entering C_7 . The complete 10^{11} audit gives the more precise basin proportions reported in Table 1. These percentages are therefore reported separately rather than mixed together.

11 The 10^{11} record for maximum steps to an attractor

The final audit found a maximum of

$$487$$

steps to a known attracting cycle, attained by

$$n = 81,078,368,921.$$

This statistic is deliberately called *steps to an attractor*. It is distinct from the first-lower stopping time $\sigma(n)$ defined in (3), and the two quantities should not be conflated.

12 The 10^{11} record for largest peak

The largest trajectory value found in the complete audit is

$$1,261,811,304,250,099,959,379,$$

obtained from

$$n = 96,110,420,480$$

at step

$$165.$$

In scientific notation the peak is approximately 1.2618×10^{21} . It exceeds the unsigned 64-bit maximum $2^{64} - 1$, which independently demonstrates the need for wider arithmetic in the extension scanner. The supplied implementation therefore promotes trajectories to 128-bit arithmetic when necessary.

13 Factor-of-3 statistics

Across the complete range through 10^{11} , the combined number of extra factors of 3 removed by compression was

$$2,755,160,788,171.$$

The largest number of extra factors removed in one step in the extension scan was

$$23,$$

attained for the starting value 70,607,384,120 at its first compressed step. These measurements give direct finite-domain information about the valuation mechanism used in the 3-adic heuristic.

14 Why the two cycles are natural candidates

The first cycle is created by a short interaction with the factor 3:

$$1 \rightarrow 2 \rightarrow 3 \rightarrow 1$$

in the uncompressed description, which becomes $1 \rightarrow 2 \rightarrow 1$ after normalization.

The second cycle contains a longer increasing chain:

$$7 \rightarrow 10 \rightarrow 14 \rightarrow 19 \rightarrow 26 \rightarrow 35 \rightarrow 47,$$

followed by the strong compression

$$47 \rightarrow 63 \rightarrow 7.$$

The cycle therefore shows exactly the type of competition suggested by the 3-adic heuristic: several steps with approximate factor $4/3$ can be balanced by a later removal of powers of 3.

The 10^{11} basin counts show that the second cycle captures the great majority of tested starts. This does not mean that C_7 is globally dominant in a mathematical sense; it means only that it is dominant in the finite computational domain tested here.

15 Computational conjecture

The numerical evidence leads to the following conjecture.

Conjecture (Two-Cycle Convergence Conjecture). Every positive integer, under the compressed ternary 3-adic map defined by (1), eventually enters one of the two cycles

$$C_1 = (1, 2)$$

and

$$C_7 = (7, 10, 14, 19, 26, 35, 47).$$

In particular, no other periodic cycle and no divergent orbit to infinity exists in the positive integers.

The exhaustive computation through 10^{11} provides strong finite evidence for this conjecture. It does not prove the conjecture for all positive integers.

16 What is still needed for a proof?

A proof of the conjecture would have to address at least three main issues.

First, the negative drift in (9) depends on a Haar-measure or uniform-residue heuristic. A rigorous proof would need to control the deterministic frequency of 3-adic valuations along every orbit, or avoid the probabilistic model entirely.

Second, a proof must rule out every other periodic cycle. The computation found only C_1 and C_7 in the tested range, but a cycle whose entries lie above the tested range is not excluded by finite computation.

Third, a proof must rule out an orbit that grows forever. The global peak of more than 10^{21} from a start below 10^{11} shows that very large temporary growth is possible. Therefore a proof cannot rely only on small numerical behavior.

These are genuine open parts of the problem as studied here. The paper therefore treats the theoretical contraction as a heuristic and the 10^{11} result as a finite theorem about the tested set, while treating the all-integers statement as a conjecture.

17 Relation to the author's earlier ternary map

The present map is related to the author's earlier paper, but it is not the same map. The earlier work studied the ternary $(4k \pm 1)/3$ map with a fixed point at 1. The present map uses the pre-reduction rule

$$G(3p + 1) = 4p + 2, \quad G(3p + 2) = 4p + 3,$$

followed by complete removal of factors of 3, and it has the two observed cycles C_1 and C_7 .

The earlier article is therefore cited as related background, not as a source of the present computational result. The two studies should be kept as separate records because their maps, terminal structures, and numerical behavior are different.

The author's previous article is available through Zenodo at [doi:10.5281/ZENODO.22195651](https://doi.org/10.5281/ZENODO.22195651), and the computational repository is github.com/FarhadBanazadeh/sequence-computations.

18 Reproducibility

The accompanying computational package contains this LaTeX source, the generated PDF, the original 10^9 audit materials, the complete $10^9 < n \leq 10^{11}$ extension summary and chunk-by-chunk checkpoint CSV, and the C sources used for the extension. The FAST extension source is the recommended reproduction program for the new range.

A representative compilation command is

```
clang -O3 -march=native -flto -DNDEBUG -std=c11 -pthread \\  
  scan_ternary_4k_plus_1_2_over_3_FAST_mac_v2.c -o scan_ternary_FAST  
./scan_ternary_FAST
```

The extension scan used four threads and a raw integer chunk width of 30,000,000. Its final summary reports 66,000,000,000 admissible starts, zero unknowns, zero overflows, zero step-limit cases, and a wall time of 30,700.091 seconds. The checkpoint CSV records all 3,300 chunks through 100,000,000,000.

The finite computation is deterministic. The aggregate 10^{11} statistics in Table 1 are obtained by adding the extension totals to the previously published 10^9 audit totals. Record maxima are taken over both ranges.

19 Conclusion

This paper introduced a new ternary 3-adic Collatz-type map in which the pre-reduced value is approximately $4n/3$ and all factors of 3 are then removed. The resulting compressed dynamics have two observed cycles, $C_1 = (1, 2)$ and $C_7 = (7, 10, 14, 19, 26, 35, 47)$.

The theoretical analysis gives a simple explanation for why contraction is plausible. Under the Haar-measure heuristic,

$$\Pr(v_3 = k) = \frac{2}{3^{k+1}}, \quad \mathbb{E}[v_3] = \frac{1}{2},$$

and hence

$$\delta = \ln \frac{4}{3} - \frac{1}{2} \ln 3 \approx -0.2616.$$

The corresponding average multiplicative factor is about 0.7698, suggesting a 23.02% decrease per step on the logarithmic scale. This is useful evidence, but it is not a proof.

The computational evidence is much stronger in the finite domain. Every one of the 66,666,666,667 admissible starting values up to 10^{11} was classified into one of the two observed cycles, with zero unknown cases. The C_1 basin contains 2,179,160,439 starts and the C_7 basin contains 64,487,506,228. The combined total is 5,449,566,633,079 steps to known attractors, with mean 81.7434994958. The maximum is 487 steps to an attractor, attained by $n = 81,078,368,921$, and the largest trajectory peak is 1,261,811,304,250,099,959,379, attained from $n = 96,110,420,480$ at step 165.

The results strongly support the Two-Cycle Convergence Conjecture: every positive integer eventually enters C_1 or C_7 , with no other cycle and no divergent orbit. The conjecture remains open. The purpose of the present paper is to provide a clear mathematical definition, a complete theoretical analysis of the 3-adic mechanism, a large finite computation, and enough source and data material for independent reproduction.

Data and code availability

The supplementary archive associated with this paper contains the original and extension verification sources, the complete chunk checkpoint, the final extension summary, the earlier audit results, the article source, and documentation of the computation. The paper source is supplied separately so that the PDF and LaTeX source can be deposited independently on Zenodo and GitHub.

References

- [1] J. C. Lagarias, “The $3x+1$ problem and its generalizations,” *The American Mathematical Monthly*, 92(1), 3–23, 1985. DOI: [10.2307/2322189](https://doi.org/10.2307/2322189).
- [2] Farhad Banazadeh, “A Ternary $(4k \pm 1)/3$ Collatz-Type Map: Computational Verification up to 10^9 ,” 2026. Zenodo DOI: [10.5281/ZENODO.22195651](https://doi.org/10.5281/ZENODO.22195651).
- [3] Farhad Banazadeh, “Sequence Computations,” GitHub repository, 2026. github.com/FarhadBanazadeh/sequence-computations.